1. Difference between "Love" and "love" in NLP

In Natural Language Processing (NLP), **"Love"** (capital L) and **"love"** (lowercase l) are treated as **different tokens** unless preprocessing is applied.

- **"Love"** may appear at the beginning of a sentence or as a proper noun (e.g., a name or title).
- **"love"** is typically used as a common noun or verb.

If we don't normalize text (e.g., converting everything to lowercase), the model may treat them as unrelated words, which can lead to **data fragmentation** and poorer performance.

2. What happens if stopwords are not removed?

Stopwords are common words like *the, is, in, and, of*.

If they are not removed:

- The dataset becomes **larger and noisier**.
- Models may give importance to **irrelevant words** instead of meaningful ones.
- Processing becomes **slower and less efficient**.
- It can reduce accuracy in tasks like classification or keyword extraction.

However, keeping stopwords is sometimes useful depending on the task.

3. Two real-world scenarios where removing stopwords can be harmful

a) Sentiment Analysis

- Example: *"I do not like this movie"*
- Removing stopwords may delete **"not"**, changing the meaning to *"I like this movie"* → completely wrong sentiment.

b) Question Answering / Search Queries

- Example: *"What is the capital of India?"*

- Removing stopwords like *"what", "is", "the"* may distort the structure and intent of the question, making it harder for systems to interpret correctly.

4. Difference between Stemming and Lemmatization

Stemming:

- Cuts words down to their root form using simple rules.
- Often produces **incomplete or incorrect words**.
- Example:
    - *running → run*
    - *studies → studi*

Lemmatization:

- Converts words to their **base or dictionary form (lemma)** using linguistic knowledge.
- Produces **real, meaningful words**.
- Example:
    - *running → run*
    - *better → good*

Key Difference:

- Stemming is **faster but less accurate**.
- Lemmatization is **slower but more accurate and meaningful**.